

Arnab Roy

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Research Interests

- Scientific Machine Learning
- Diffusion Models and Simulation-Based Inference
- Operator Learning and Neural Operators
- Inverse Problems and Parameter Estimation for PDEs and ODEs
- Computational Modeling of Dynamical Systems
- Numerical Analysis of Differential Equations

Education

University of Delaware, Newark, DE, USA

August 2022 – Expected June 2027

PhD in Mathematics, Department of Mathematical Sciences

Advisor: Dr. Tobin A. Driscoll

GPA: 3.746/4.000

Indian Institute of Science Education and Research (IISER) Berhampur, India

August 2016 – May 2021

BS-MS Dual Degree (Major: Mathematics, Minor: Physics)

CPI: 9.44/10.00

Advisor: Dr. Ritabrata Sengupta

Thesis: Interpolation by Different Types of Quantum Channels Using Conic Programs and Further Generalisation

Publications and Preprints

Journal Articles

Chen, Q., **Roy, A.**, and Driscoll, T. A. Operator learning for models of tear film breakup. Mathematical Medicine and Biology: A Journal of the IMA, 2026. Advance access, article dqag009.

DOI: <https://doi.org/10.1093/imammb/dqag009> | arXiv: [2601.08001](https://arxiv.org/abs/2601.08001)

Preprints

Roy, A. and Patra, S. Interpolation by Different Types of Quantum Channels Using Conic Programs and Further Generalisation. arXiv: 2104.07254, 2021.

DOI: <https://doi.org/10.48550/arXiv.2104.07254>

Research Experience

Stochastic Operator Learning for Tear Film Dynamics Using Diffusion Models

2025 – Present

University of Delaware, PhD Research under Dr. Tobin A. Driscoll

- Developing diffusion-based simulation-based inference models to learn posterior distributions of parameters governing tear film ODE systems from observed fluorescence intensity data.
- Implementing conditional diffusion models using BayesFlow and PyTorch for probabilistic parameter inference.

GitHub: [Tearfilm_SBI_diffusion_ODE](https://github.com/ArnabRoy/SBI_diffusion_ODE)

Diffusion-Based Forecasting for Kuramoto–Sivashinsky PDE

2025 – Present

University of Delaware, PhD Research

- Developing diffusion models to generate samples from the attractor distribution of the Kuramoto–Sivashinsky equation.

- Using learned representations as latent state spaces for downstream forecasting models.

GitHub: [Diffusion_forecast_KS](#)

PINN-Based Surrogate Modeling for Tear Film PDE Systems

2025 – Present

University of Delaware, PhD Research

- Developing physics-informed neural networks as surrogate solvers for tear film PDE systems.
- Learning mappings from model parameters to spatiotemporal solution fields.

GitHub: [Tear_Film_PDE_Forward_surrogate](#)

Operator Learning for Models of Tear Film Breakup

2023 – 2025

University of Delaware, PhD Research under Dr. Tobin A. Driscoll

- Developed neural operator models to learn mappings from fluorescence intensity to tear film thickness and osmolarity.
- Implemented Fully-connected Neural Nets, Fourier Neural Nets, PCA-Neural Nets and related architectures as surrogate models for dynamical systems governed by differential equations.
- Published in Mathematical Medicine and Biology: A Journal of the IMA (2026).

Paper: doi.org/10.1093/imammb/dqag009 | GitHub: [sciml-tearfilms](#)

Interpolation by Quantum Channels Using Conic Programming

2020 – 2021

IISER Berhampur, Undergraduate Thesis under Dr. Ritabrata Sengupta

- Developed optimization-based methods to determine existence and construction of quantum channels interpolating prescribed input-output quantum states.
- Applied conic programming techniques to characterize feasible interpolation operators.

Paper: doi.org/10.48550/arXiv.2104.07254

Teaching Experience

Instructor of Record — Department of Mathematical Sciences, University of Delaware

MATH 351: Engineering Mathematics I Fall 2026 (34 students)

MATH 351: Engineering Mathematics I Summer 2026 (8 students)

MATH 305: Applied Mathematics for Biomedical Engineering Spring 2026 (~38 students)
Applied ordinary differential equations for biomedical engineering students, covering integrating factors, systems of ODEs, and second-order constant coefficient equations.

MATH 243: Analytic Geometry and Calculus C Fall 2025 (~80 students)

MATH 242: Analytic Geometry and Calculus B Summer 2025 (~30 students)

MATH 243: Analytic Geometry and Calculus C Summer 2024 (~30 students)

MATH 242: Analytic Geometry and Calculus B Winter 2024, two sections (~30 each)

Responsibilities included delivering lectures, designing assignments and exams, and assessing student performance.

Teaching Assistant — Department of Mathematical Sciences, University of Delaware

MATH 637: Mathematical Techniques in Data Science Fall 2024, Spring 2024

MATH 219: Data Science I Spring 2024

MATH 243: Analytic Geometry and Calculus C Fall 2023

MATH 242: Analytic Geometry and Calculus B Spring 2023

Responsibilities included conducting discussion sessions, grading coursework, and holding office hours.

Technical Skills

Programming Languages: Python (Proficient), Julia (Intermediate)

Machine Learning and Scientific Computing: PyTorch, TensorFlow, Keras, DeepXDE, BayesFlow, Scikit-learn, NumPy, SciPy

Scientific Machine Learning and Numerical Methods: Neural Operators (DeepONet, Fourier Neural Operators), Diffusion Models, Physics-Informed Neural Networks (PINNs), Neural ODEs, Transformers, Fourier Neural Networks, Finite Element Methods, Optimization, Numerical Solution of Differential Equations

Computing and Development Tools: Git, Linux, DARWIN HPC Cluster, VS Code, Anaconda, Jupyter Notebook

Data Formats and Workflow Tools: HDF5, YAML, SLURM

Honors and Awards

Institution Silver Medal — IISER Berhampur 2021

Awarded for highest cumulative grade point index in the Department of Mathematical Sciences.

INSPIRE Fellowship — Government of India 2016 – 2021

National merit-based fellowship awarded to the top 1% of students for pursuing higher education in science.

Conferences and Workshops

SIAM Annual Meeting (AN26), Cleveland, OH July 2026

Presentation: Can a single neural network generate samples from an entire family of attractors — from steady to fully chaotic?

Mid-Atlantic Numerical Analysis Day, Temple University November 2025

Presentation: Operator Learning for Tear Film Dynamics

Graduate Student Mathematical Modeling Camp (GSMMC), University of Delaware June 2023

Mathematical Problems in Industry (MPI), New Jersey Institute of Technology June 2023

Languages

Bengali — Native | English — Fluent | Hindi — Conversational

References

References available upon request.